

Script of the Design Course

Design Course: RF- and Microwave Circuits and Systems

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Version 12.0



EEH, Fritz-Hüttinger chair energy-efficient
high-frequency electronics
Freiburg University

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Introduction

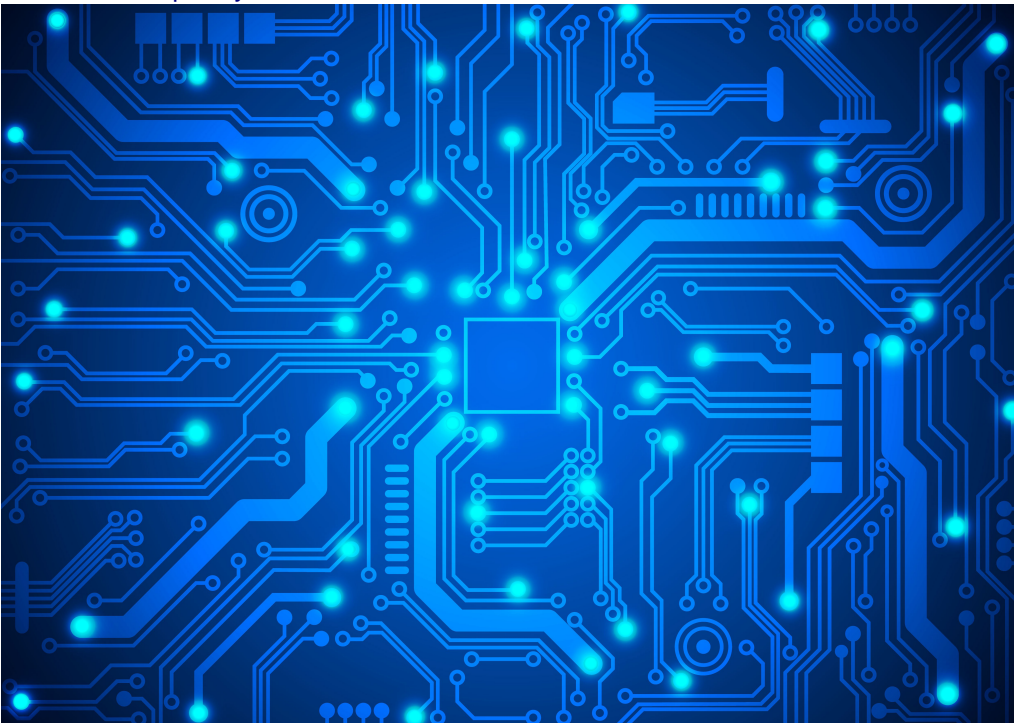


Introduction



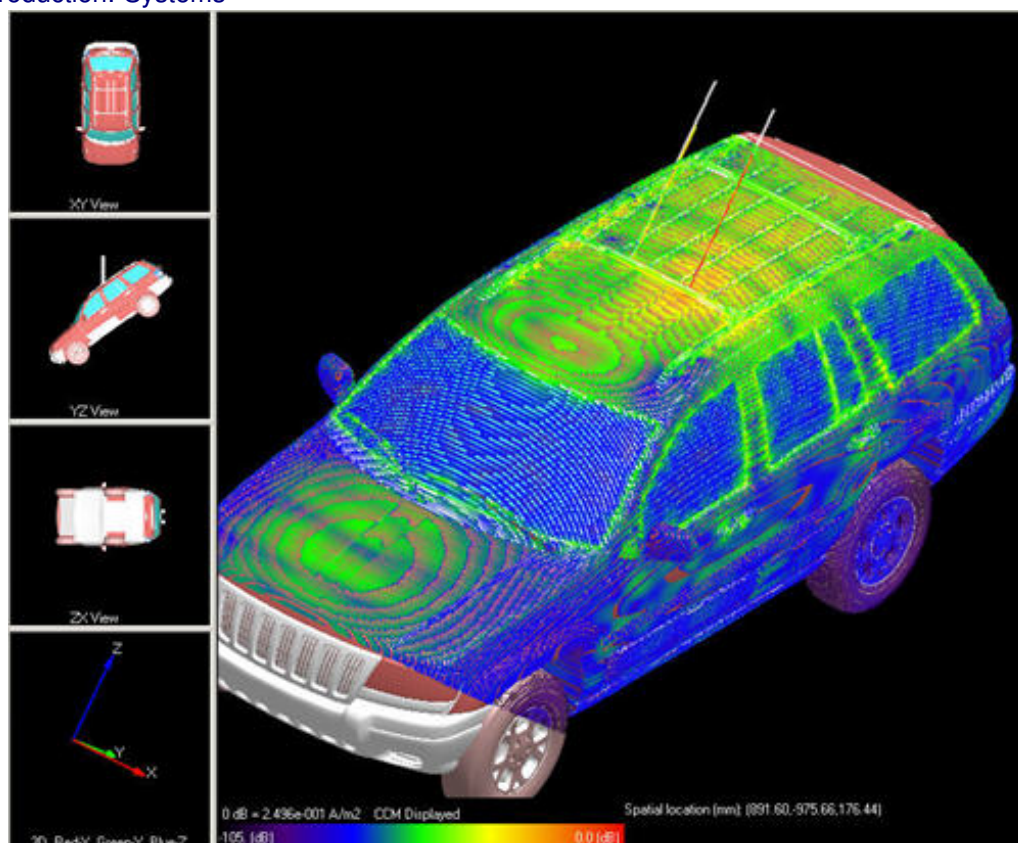
1.4

Introduction: Complexity



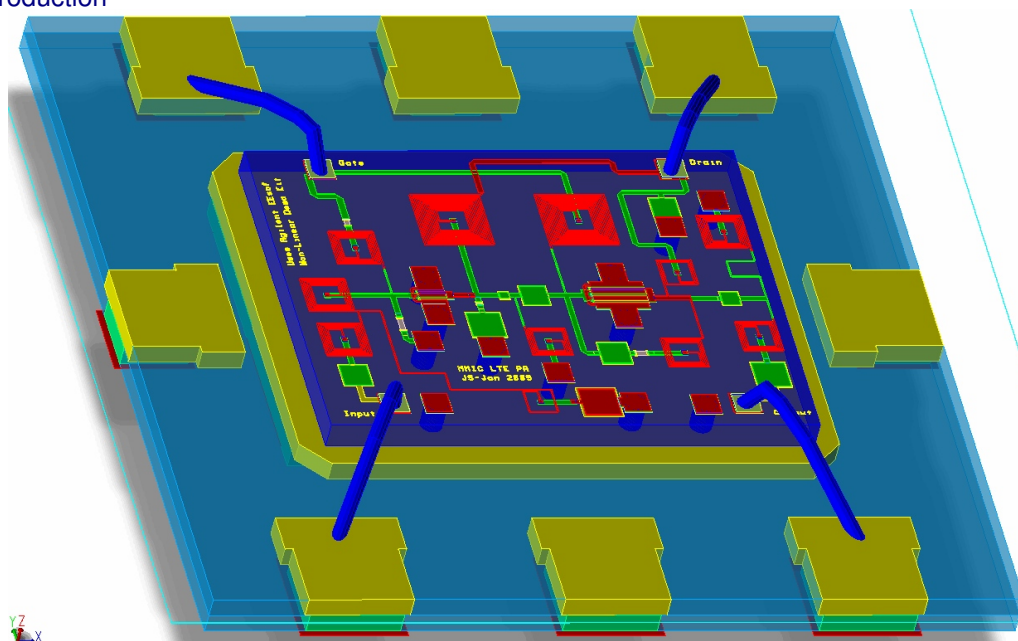
1.5

Introduction: Systems



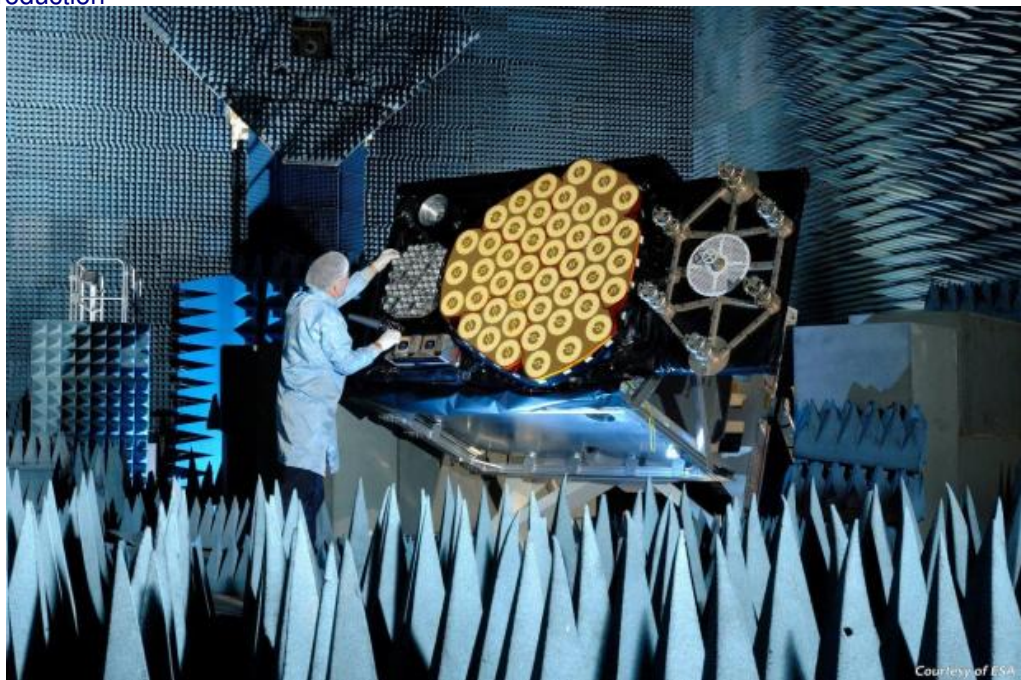
1.6

Introduction



!!!! Frequency domain !!!! vs. Cadence Virtuoso: Time domain

1.7



Dates with exercise

- Mo 1.) 20.04.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 2.) 27.04.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 3.) 4.05.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 4.) 11.05.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 5. 18.05.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 6. 25.05.2026 no lecture, pentacost break SR 00 014 (G.-Köhler-Allee 078)
- Mo 7. 1.06.2026 live SR 00 014 (G.-Köhler-Allee 078)

1.9

Dates with exercise II

- Mo 8. 8.6.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 9. 15.06.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 10. 22.06.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 11. 29.06.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 12. 6.07.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 12. 13.07.2026 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)
- Mo 13. 20.07.2026 Presentation of final results from competition 16:00 - 17:00 SR 00 014 (G.-Köhler-Allee 078)

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Date and Location

Dates with exercise

1. Mo, 20.4.2026: 16-17.00 h Introduction and group formation
2. Block 1: Mo., 27.4.2026: (issue block 1a on ILIAS)
3. Block 1: Mo., 4.5.2026: (issue block 1b on ILIAS)
4. Block 2: Mo., 11.5.2026 (issue block 2a on ILIAS), (Deadl. Block 1)
5. Block 2: Mo., 18.5.2026 (issue block 2b on ILIAS)
6. Block 3: Mo., 1.6.2026 (deadline block 2 on ILIAS), (issue block 3a on ILIAS)
7. Block 3: Mo., 08.6.2026 (issue block 3b on ILIAS)
8. Block 4: Mo., 15.6.2026 (Deadline block 3), (issue block 4a on ILIAS)
9. Block 4: Mo., 22.6.2026, (Issue Block 4b, COMPETITION !!!)
10. Block 5: Mo. 29.6.2026 (deadline block 4), (issue block 5a on ILIAS)
11. Block 5: Mo., 06.7.2026 : (issue block 5b ON ILIAS)
12. Mo., 13.7.2026, Final presentations based on 4b (deadline block 5 on ILIAS)

1.11

Rules of the Exercises

- 5 blocks Exercises (every one/two weeks) with overall max. 10 tasks in a given format...
- First: starting 27.4.2026... (two weeks to do, sometimes more)
- 4 blocks to be completed successfully (with 50% correct) (three people per group)
- Voluntary Competition: Final presentation (5 mins) of the group to the plenary based on a task
- Exercises are given out on ILIAS and also submitted there.
- **Timing is day of the deadline, 16 h**

1.12

1 Abstract

The students will be enabled to understand, design and layout modern RF- and microwave components and systems by means of the electronic design environment Advanced Design System including the two- and three dimensional electromagnetic simulators Momentum and EMPro 3D. The detailed use of a complex RF-software environment is a dedicated target of this course. This includes the numerical analysis of complex passive and active devices, the design and layout of hybrid and integrated circuits, and their packaging and signal flow. The students will be competent to design and layout passive and active RF-structures including packages and interconnects and circuits of relevance to everyday communication and sensing. The competence includes in-depth understanding and treatment of complex microwave systems and of general system design including the treatment of complex modulated signal flows.

Introduction

Materials and Script

The material is provided at the beginning and during the lecture.

Email: ruediger.quay@inatech.uni-freiburg.de

- Foils: Intro materials from Keysight
- ADS Video Library on www.youtube.com
- Script (supporting materials)
- Exercises

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Videos

1. Watch ADS videos on youtube using ADS : www.youtube.com , Keyword: ADS Video Library
2. Basics and setup:
3. <https://www.youtube.com/watch?v=04BfajzXnmM&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz>
4. Understanding Workspaces:
5. <https://www.youtube.com/watch?v=7jNm400BaiE&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz&index=2>
6. The basics of schematics:
7. <https://www.youtube.com/watch?v=wqyyOCzxGS0&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz&index=4>
8. Data displays:
9. <https://www.youtube.com/watch?v=jIg10RihLQk&list=PLtq84kH8xZ9HNMCB7xKGXrciiEqqHNQGz&index=6>

1.14

WWW Support 1

ILIAS Support

<https://ilias.uni-freiburg.de>

search for: Design Course: RF- and Microwave Systems

password: design2026

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WWW Support 2

This exercise is recorded, attendance is, however, required



1.16

Windows Access to the Uni Network

- Get login at university, get additional TF login
- Install VPN from University server
- use ssh from linux
- If you start from Windows:
- Use, e.g., xming as windows X-server (for export of the display)
- Install Putty ssh client on windows

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Form groups of Three

- The exercises are submitted from a group of three people
- Find two partners !!!
- Send your group to ruediger.quay@inatech.uni-freiburg.de, including TF login name.
- Exercises start april 27th 2026 !

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Install an ADS Version on a local computer

Last Updated: 05 March 2020

ADS - Windows® Operating Systems	
Product	Windows
ADS 2020 / ADS 2020 Update 1.0 / ADS 2020 Update 2.0 64-bit Operating Systems	Windows 10 (64-bit) Windows 7 Enterprise (64-bit)
ADS 2019/ ADS 2019 Update 1.0 64-bit Operating Systems	Windows 10 (64-bit) Windows 7 Enterprise (64-bit)
ADS 2017/ ADS 2017 Update 1.0 64-bit Operating Systems	Windows 10 (64-bit) Windows 8 Enterprise (64-bit) Windows 7 Enterprise (64-bit)
ADS 2016.01 64-bit Operating Systems	Windows 10 (64-bit)* Windows 8 Enterprise (64-bit) Windows 7 Enterprise (64-bit)
ADS 2015.01 64-bit Operating Systems	Windows 8 Enterprise (64-bit) Windows 7 Enterprise (64-bit)

You can work on the aragorn but local has proven to be easier...

Organize your data in a good structure

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Login to Application Server for the license

From Remote: Create RAS Access in Mylogin

Go into University net with VPN: check VPN Uni Freiburg

In case you use Windows to do that: use putty and install xming as windows X-server

Application Server: `aragorn.imtek.uni-freiburg.de`, `ssh-X-1 (yourlogin) aragorn.imtek.uni-freiburg.de`

Shells usable: `bash` and `tcsh`

Start: `ads` after using `ssh` to the server

Licence Server will be asked: `27010(at)frodo.imtek.uni-freiburg.de`

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Backup Server !

Application Server: `samweis.imtek.uni-freiburg.de` (only via VPN)

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Recommendation

Use Licence server only, install ADS on your local device ! Use ADS 2023 Update 1.0

1.22

The following references may be useful to provide more detailed information.

References

- List of useful ADS Videos
- Keysight Software Support: <http://www.keysight.com/>
- Introduction to ADS, ADS_Basics.pdf (ILIAS)
- Momentum Momentum_Basics.pdf (ILIAS)

1.23

2 Large-Signal: Educational Targets

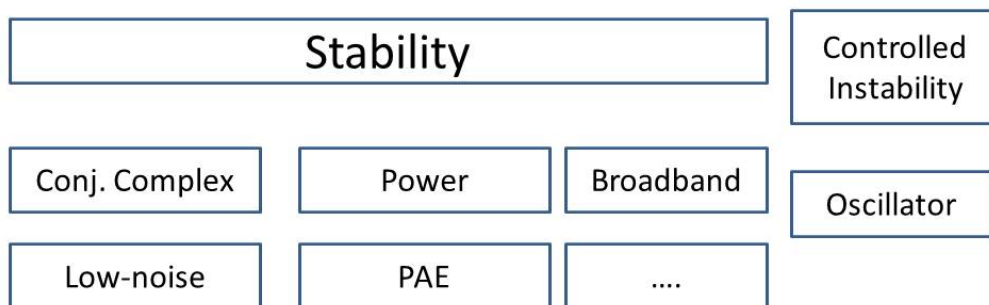
Educational targets

- Understand matching choices
- Understand the large-signal situation beyond an input and output load of 50 Ohms
- Perform a loadpull analysis and understand the trade-offs
- Understand and use the term efficiency of PAE and DE

8.25

Design choices

Matching choices



8.26

3 Power Amplifiers

3.1 Designing a Power Amplifier

Loadpull Simulation

The screenshot shows the ADS 2012 Help application window. The address bar displays the URL: `qthelp://ads.2012.08.app/doc/examples/Load-Pull_Simulations.html`. The left sidebar contains a tree view with the following structure:

- Getting Started
 - Installation and License Se...
 - ADS Overview
 - Upgrading from ADS 2009...
 - Customization and Configu...
 - ADS 2012 Release Notes
- Applications
 - Design Guides
 - Design Kits
 - Examples
 - Application Examples
 - A-to-D D-to-A Appli...
 - Budget Analysis Ap...
 - Load-Pull Simulations**
 - Radar Applications ...
 - Signal Integrity Appl...
 - VPI ADS Link
 - Wireline Applications
 - Behavioral Model Exam...
 - Communication System...

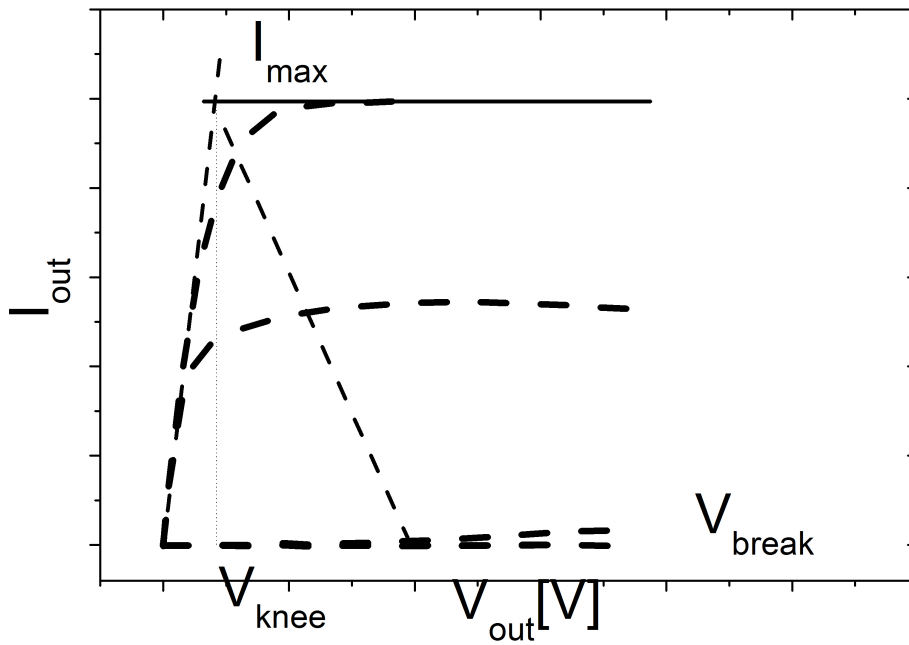
The main content area displays the title "Load-Pull Simulations" and a breadcrumb trail: `Home > Applications > Examples > Application Examples`. Below the title, there is a section "On this page" with links to [Objective](#), [Setup](#), [Analysis](#), and [Notes](#). A paragraph states: "The Load Pull Simulation Guide is available in the schematic window under the **DesignGuide** menu." Below this is the "Objective" section, which explains that load-pull simulations generate contours for load impedances and power. A link [BACK TO TOP](#) is provided. The "Setup" section contains a numbered list of three items:

1. "HB1Tone_LoadPull" is a simulation set-up that generates actual contour lines for output power and power-added efficiency.
2. "HB2Tone_LoadPull" is identical to "HB1Tone_LoadPull" except that it shows all the equations, and includes some explanatory text.
3. "HB1Tone_LoadPull.dds" displays the output power and power-added efficiency contours

3.1.1 Recall Power Amplifier Design

Class-A/B Conditions: PAE

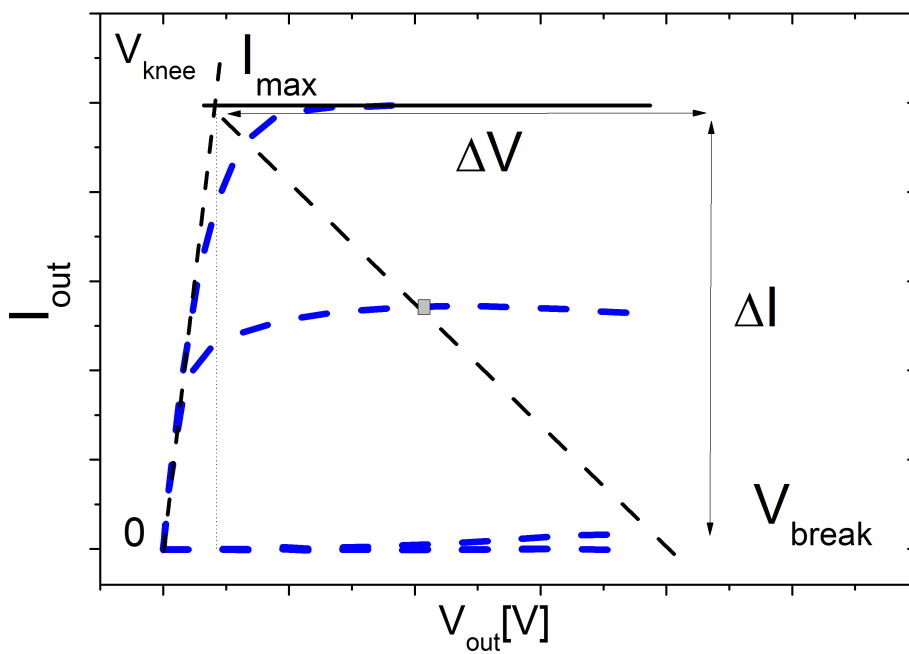
Loadline choices



8.28

Class-A Conditions: Power

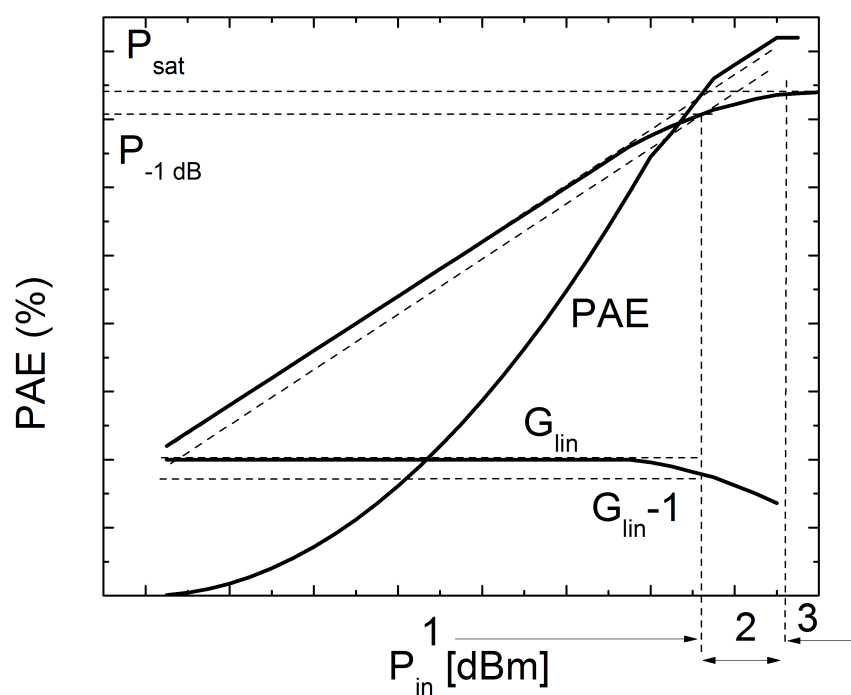
Loadline



8.29

Power compression: Example

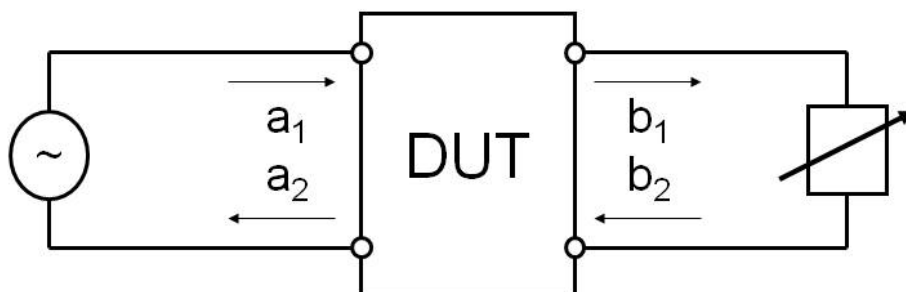
Power compression



8.30

Loadpull

How to tune ?



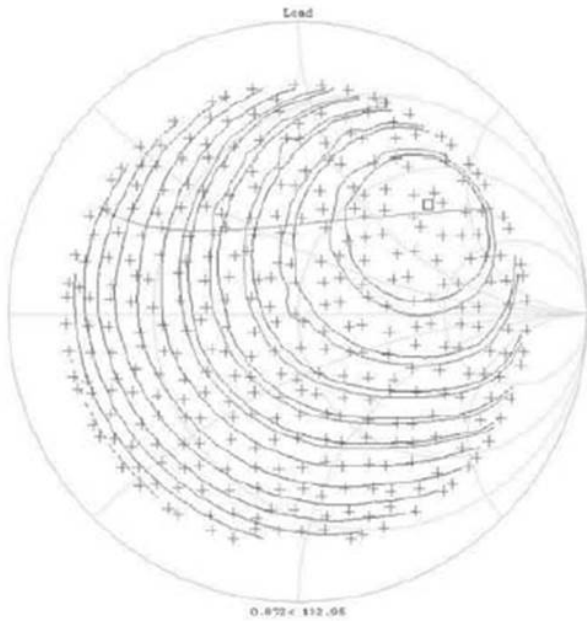
Typically first at output at f_0 , but also at input possible...

8.31

Loadpull Technique

Load pull

crosses: data points

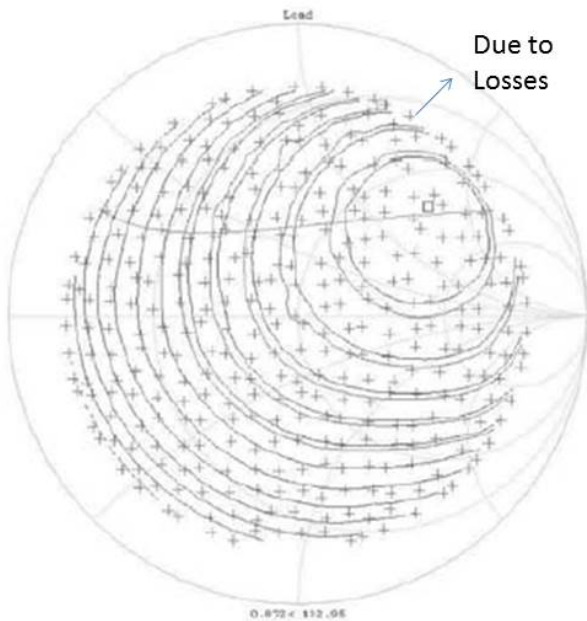


Red: Isolines of PAE Black: Isolines of power

Loadpull Technique 2

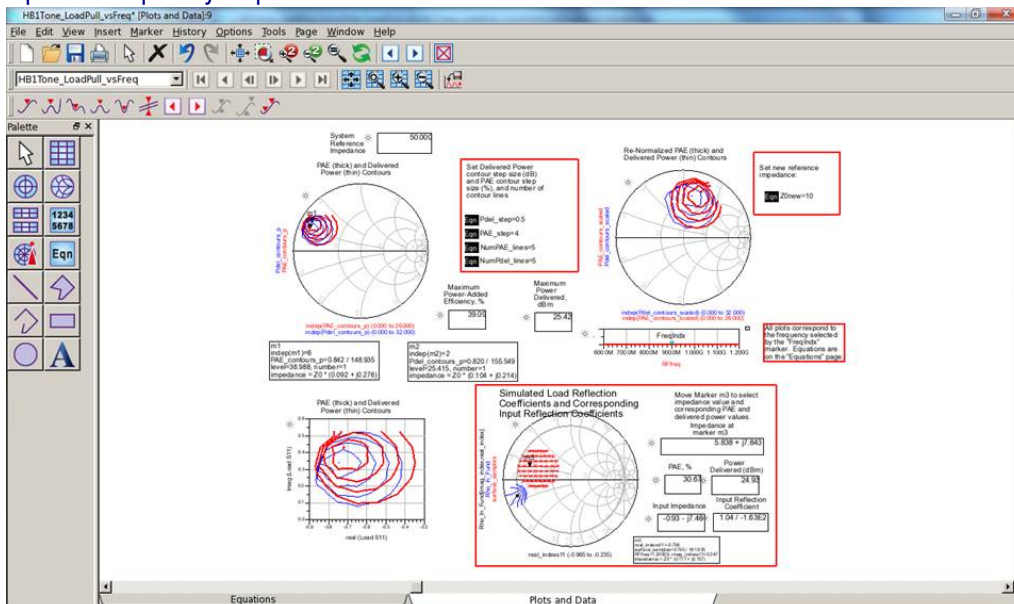
Load pull

crosses_: data points



Red: Isolines of PAE Black: Isolines of power

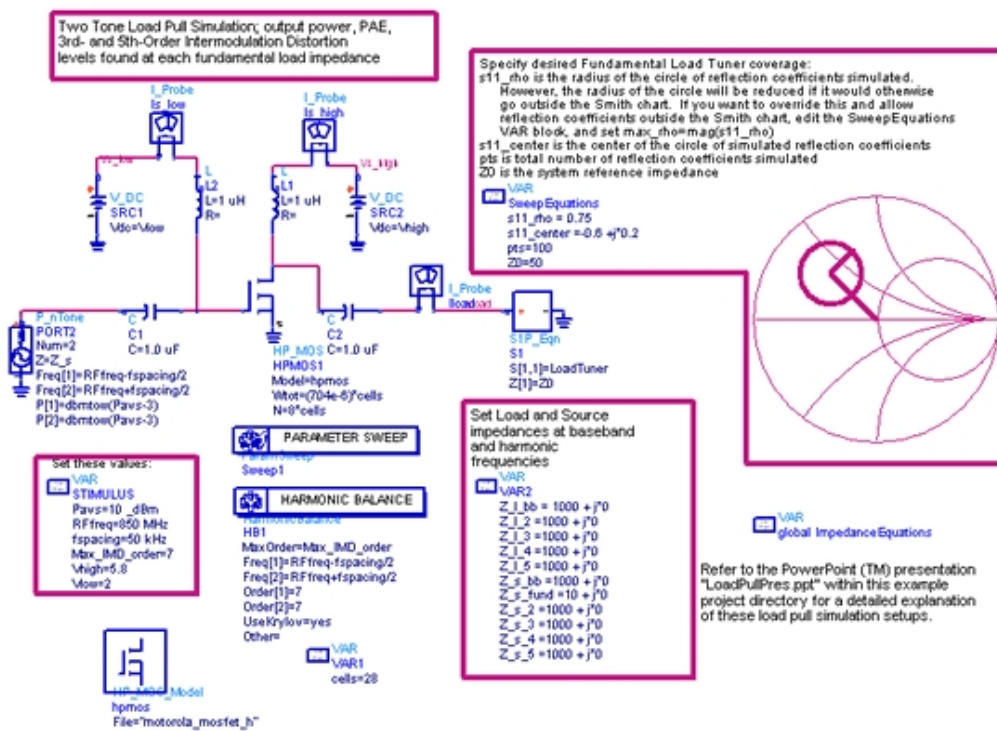
Loadpull is frequency-dependent



8.34

Loadpull Simulation: Example

Load pull example



8.35

4 Task 14: Loadpull

4.1 Tasks 14

Tasks and Exercises 14 Block 4 part 1

1. Explanation: Open the example `examples\RFboard\loadpullwrk`
2. Explanation: Relax and concentrate on the `..ITone..` examples
3. Explanation: This is the large-signal simulation of a former Motorola, now Qualcomm, power MOSFET at a frequency of around 1000 MHz.
4. Explanation: Go to the setup `HB1Tone_LoadPull`.
5. Task: First of all, create a well known small signal-situation for very low input powers in this large-signal set-up. Set V_{high} to 6.5 V. at 1000 MHz, plot the P_{del} (delivered power) contours of the power in the Smith Chart for $V_{low} = 2$ V and 3 V (5 points).
6. Task: Explain what you change with V_{low} . Use the term: amplifier classes (10 points)

8.36

Tasks and Exercises 14 Block 4 part 1

7. Task: For $V_{high} = 6.5$ V and $V_{low} = 3$ V: Change the number of MOS cells from 28 to 56 (times two). Also increase P_{avs} by 3 dB accordingly and 7 dBm for 28 and 10 dBm for 56 cells. What do you expect to happen with the magnitude of the input impedance simulated. Think simple. What happens with the impedance of a capacitor if you put two of them in parallel ? (5 points)
8. Task: Looking at the output now: Put P_{avs} to 10 dBm and the number of cells back to 28. V_{low} is still 3 V, $V_{high} = 6.5$ V. Set V_{high} to 9 V instead of 6.5 V, repeat the simulation at $P_{avs} = 10$ dBm and compare/explain what changes with real part of the output impedance as a function of bias. Also explain changes in output power and gain (10 points)
9. Task: Explain the term PAE (power added efficiency) and drain efficiency (DE) (5 points).
10. Explanation: Goto the setup `HB1ToneLoadPullSweep`.

8.37

Tasks and Exercises 14 Block 4 part 1

12. Task: Determine the gain compression curves for $V_{high} = 6.5$ V and 9 V for the input power range given at 1000 MHz as given in the default in the power sweep template using the $P_{avs} = 7$ dBm for the LP ? Please show the full input power sweep of gain, power, and PAE for both voltages at 1000 MHz frequency for the setting of the max. PAE load (5 points)
13. Are you using the same impedance for both voltages of V_{high} ? Why (not)? (think about the output characteristics and the real part of the load). (Two plots and three sentences) (10 points)

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Formal: Tasks 14 Block 4 part 1: 50 points

- 14.5 Give the two plots for the two bias setting of V_{low} (5 points)
- 14.6 Give five sentences and use the term Class-B and Class-A/B (10 points)
- 14.7 Three sentences on the behavior and two on the parallel capacitors (5 points)
- 14.8 Max. 5 sentences on the behavior and two plots on loadpull, one for each bias (10 points)
- 14.9 Two equations and five sentences max. (5 points)
- 14.12 Two plots (5 points)
- 14.13: Two plots and three sentences (10 points)

8.39